LEARNER GUIDE

Expanding and factorising

Algebraic Expressions and Identities · Lesson 2. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

One rectangle can be described two ways: as the sum of its parts, or as length × width. **Expanding** turns a product into a sum: $(2x + 3)(x – 4) = 2x^2 – 5x – 12$. **Factorising** turns a sum back into a product. Take out the highest common factor first: $6x^2 + 9x = 3x(2x + 3)$. To factorise $x^2 + bx + c$, find two numbers that multiply to c and add to b.

What you are learning to notice

- In the grid for $(x + p)(x + q)$, the last cell is pq and the two middle cells add to $(p + q)x$.
- If the number term is negative, one of the two numbers is negative. If it is positive but the x coefficient is negative, both are negative.
- Difference of two squares: $x^2 – 25 = (x + 5)(x – 5)$. The two middle cells cancel.
- Fully factorised means nothing more can be taken out: $2x(4x + 6)$ is not finished, but $4x(2x + 3)$ is.
- Always check a factorisation by expanding it.

Moving through it, one step at a time

To expand, draw the grid: one term of each bracket along each edge. Fill every cell, then collect like terms. To factorise, first take out the highest common factor. Then, for $x^2 + bx + c$, list the factor pairs of c (with signs), find the pair that adds to b, and write the two brackets. Expand to check.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Expanding with the grid

Expand and simplify $(2x + 3)(x - 4)$.

$(2x + 3)(x - 4)$

×	x	-4
$2x$	$2x^2$	$-8x$
3	$3x$	-12

1. Put one bracket down the side and the other as the column headings.	2 by 2 grid
2. Fill each cell: its row term × its column term, keeping the signs.	$2x^2$, $-8x$, $3x$, -12
3. Collect the x terms (the shaded cells).	$-8x + 3x = -5x$
4. Write the sum.	$2x^2 - 5x - 12$

$(2x + 3)(x - 4) = 2x^2 - 5x - 12$.

Example 2 — Factorising: find the sides

Factorise $x^2 - x - 12$.

The rectangle for $(x + 3)(x - 4)$

×	x	-4
x	x^2	$-4x$
3	$3x$	-12

1. The last cell must be -12, and the two shaded cells must add to $-x$.	multiply to -12, add to -1
2. Try the factor pairs of -12: 3 and -4.	$3 \times (-4) = -12$, $3 + (-4) = -1$
3. These are the sides of the rectangle.	$(x + 3)(x - 4)$
4. Check by expanding: the grid gives $x^2 - 4x + 3x - 12$.	$= x^2 - x - 12$ ✓

$x^2 - x - 12 = (x + 3)(x - 4)$.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (expanding a bracket, factor pairs, and multiplying negative numbers), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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